

Design of Experiments (DOE) Framework

Supply Chain Optimization Across Product Lines

Hollister Incorporated

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Solution Architecture Deliverable

Executive Summary

This document presents a structured Design of Experiments (DOE) framework for optimizing Hollister Incorporated's supply chain across its four major product lines: Ostomy Care, Continence Care, Wound Care, and Critical Care. The framework identifies key experimental factors, appropriate statistical methods, response variables, and projected business impacts for each product line, along with cross-cutting supply chain network experiments.

Hollister operates a global manufacturing and distribution network spanning facilities in Stuarts Draft (Virginia), Kirksville (Missouri), Ballina (Ireland), Kaunas (Lithuania), Bawal (India), and Oud Gastel (Netherlands), supported by SAP ERP infrastructure and the Hollister Production System (HPS) lean methodology. This DOE framework is designed to integrate with HPS principles and SAP-driven planning parameters to deliver measurable improvements in product quality, process efficiency, inventory optimization, and patient outcomes.

Total experiments identified: 37 DOEs across 5 domains, organized by implementation priority and regulatory complexity.

Scope and Methodology

The DOE framework applies systematic experimental design principles to both product-level manufacturing processes and enterprise supply chain operations. Each experiment is characterized by five dimensions:

1. Experimental Factors – the controllable input variables being tested
2. Levels – the specific values or conditions for each factor
3. Response Variables – the measurable outputs indicating performance
4. DOE Method – the statistical design appropriate to the experimental structure
5. Business Impact – the strategic value proposition driving the experiment

Methods range from classical full and fractional factorial designs through response surface methodology (RSM), Taguchi robust parameter design, and advanced techniques including stochastic simulation-based DOE for network-level optimization under demand uncertainty.

Hollister Product Lines and Manufacturing Footprint

Understanding the manufacturing context is essential for proper DOE scoping. Hollister's product portfolio and manufacturing network create distinct experimental domains:

- **Ostomy Care** (Hollister Ostomy + Dansac Ostomy brands) – One-piece and two-piece pouching systems, CeraPlus skin barriers with ceramide technology, accessories. Primary manufacturing: Stuarts Draft, Ballina, Kaunas.
- **Continence Care** – VaPro and Advance Plus intermittent catheters, Infyna Chic (women's catheter), male external catheters, urine collectors, leg bags. R&D center: Ballina. Manufacturing: Ballina, Stuarts Draft, Bawal.
- **Wound Care** – Hydrofera Blue antibacterial foam, Restore dressings (foam, silver, alginate, hydrocolloid), Endoform dermal template, wound cleansers. Manufacturing: Stuarts Draft.

- **Critical Care** – Tube fasteners, tube attachment devices, fecal management systems. Manufacturing: distributed across US facilities.

DOE Experiments by Product Line

1. Ostomy Care DOEs

Focus: Skin barrier performance, pouch integrity, manufacturing process parameters, CeraPlus ceramide optimization

Experimental Factor	Levels	Response Variable	Method	Business Impact
Skin barrier formulation blend ratio	3 blend ratios (ceramide concentration)	Barrier wear time (hours), skin irritation score	Full factorial 3 ²	Reduced returns, improved patient satisfaction
Pouch film laminate thickness	2 levels (current, -10%)	Odor containment rating, puncture resistance	2-factor factorial	Material cost reduction while maintaining quality
Adhesive cure time vs. temperature	3 temps × 3 cure times	Peel strength (N/cm), skin barrier integrity	Response surface (CCD)	Process throughput optimization
Flange coupling mechanism tolerance	3 tolerance bands	Audible click rate, leak frequency	Fractional factorial 2 ³⁻¹	Field failure reduction, patient confidence
Sterilization method (EtO vs. gamma)	2 methods × 3 dose levels	Bioburden, material degradation index	Split-plot design	Regulatory compliance, shelf life extension
Packaging seal width	3 widths (mm)	Sterile barrier integrity, peel force	Randomized block	Packaging waste reduction, ease of opening
CeraPlus ceramide loading concentration	4 loading levels	Moisture vapor transmission rate, skin pH	Taguchi L9 array	Peristomal skin health outcomes
Cut-to-fit vs. pre-cut stoma opening	2 types × 4 stoma sizes	Fit accuracy, application time	Nested factorial	SKU rationalization, inventory reduction

2. Continence Care DOEs

Focus: Catheter coating and design, infection prevention, patient experience, self-catheterization success

Experimental Factor	Levels	Response Variable	Method	Business Impact
Catheter hydrophilic coating thickness	3 thickness levels	Friction coefficient, UTI incidence rate	Full factorial	Clinical outcome improvement, market differentiation
VaPro protective tip geometry	4 tip designs	Bacterial transfer rate, insertion force	Balanced incomplete block	Infection risk reduction
Lubrication activation method	Water vs. saline vs. pre-activated	Lubricity retention (minutes), patient comfort score	Latin square design	User experience optimization
Catheter shaft material (PVC vs. POBE)	2 materials × 3 durometers	Kink resistance, insertion ease score	2 ³ factorial	Material cost, patient preference
Leg bag valve mechanism	3 valve types	Flow rate, leak rate, one-hand operability	Randomized complete block	Patient independence, caregiver burden
Packaging sterile field design	3 configurations	Contamination rate, time to catheterize	Crossover design	Self-catheterization success rate
Infyna Chic form factor dimensions	3 length × 2 diameter	Discreetness rating, flow rate	Response surface	Female patient adoption rate

3. Wound Care DOEs

Focus: Dressing performance, antimicrobial efficacy, ECM processing, moisture management

Experimental Factor	Levels	Response Variable	Method	Business Impact
Hydrofera Blue foam pore size	3 pore size distributions	Absorption rate (g/cm ²), antimicrobial efficacy	Full factorial 3 ²	Wound healing time reduction
Silver dressing Ag+ release kinetics	4 concentration × 2 release profiles	Zone of inhibition, cytotoxicity index	Fractional factorial	Bioburden management effectiveness
Endoform ECM processing temperature	3 temperatures	Collagen integrity, epithelialization rate	Randomized block	Wound closure outcomes
Alginate dressing fiber blend ratio	3 Ca/Na alginate ratios	Gel blocking, fluid handling capacity	Mixture design (simplex)	Exudate management optimization
Hydrocolloid adhesion vs. moisture handling	3 adhesive × 3 hydrocolloid levels	Wear time, MVTR, skin maceration score	Response surface (Box-Behnken)	Extended wear capability
Wound cleanser surfactant concentration	4 concentration levels	Biofilm disruption %, tissue viability	Dose-response design	Debridement efficacy vs. tissue safety

4. Critical Care DOEs

Focus: Securement reliability, pressure injury prevention, ICU efficiency

Experimental Factor	Levels	Response Variable	Method	Business Impact
Tube fastener adhesive formulation	3 adhesive types	Securement force, skin strip rate	Randomized complete block	Unplanned extubation reduction
Tube attachment device material	2 materials × 2 thicknesses	Flexibility score, skin pressure index	2 ² full factorial	Pressure injury prevention
Fecal collector seal geometry	3 seal profiles	Wear time, leak incidence, peristomal skin score	Balanced incomplete block	ICU nursing efficiency
Securement device repositionability	3 adhesive release profiles	Repositioning cycles before failure, residue score	Accelerated life test design	Cost per patient day reduction

5. Cross-Cutting Supply Chain Network DOEs

Focus: Network optimization, inventory buffers, ERP parameter tuning, lean kanban calibration, logistics

Experimental Factor	Levels	Response Variable	Method	Business Impact
Supplier lead time variability (raw materials)	3 supplier tiers × 2 regions	On-time delivery %, buffer stock days	Split-plot (region = whole plot)	Working capital optimization
Production batch size vs. changeover frequency	4 batch sizes × product family	OEE, inventory carrying cost, fill rate	Response surface	WIP efficiency, right-sized inventory buffers
Distribution center routing (Stuarts Draft vs. Ballina vs. Kaunas)	3 DCs × 4 market regions	Transit time, cost per unit, carbon footprint	Fractional factorial	Logistics cost and sustainability optimization
Demand forecasting model selection	3 models (ARIMA, ML, hybrid)	MAPE, bias, forecast value added	Paired comparison	Service level vs. inventory tradeoff
Sterilization queue scheduling	3 scheduling algorithms	Queue wait time, equipment utilization, throughput	Discrete event simulation + DOE	Bottleneck elimination
Kanban signal calibration (Ballina lean)	3 kanban quantities × 3 replenishment frequencies	Stockout rate, WIP turns, floor space utilization	Taguchi robust design	Lean maturity advancement
Multi-site capacity allocation	Demand scenarios × 3 allocation strategies	Total cost to serve, resilience index	Stochastic DOE (Monte Carlo)	Network resilience, risk mitigation

SAP ERP replenishment parameter tuning	Safety stock × reorder point × lot size	Service level, excess inventory \$, expedite frequency	Definitive screening design	ERP-driven inventory optimization
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Implementation Prioritization Matrix

The following matrix provides a recommended sequencing based on business impact, experimental complexity, regulatory burden, and alignment with Hollister’s strategic priorities including the €80M Ballina R&D investment and HPS lean maturity goals.

Product Line	Business Impact	Complexity	Regulatory Load	Priority	Timeline	Investment
Ostomy Care	High	Medium	High	1	Q1-Q2	\$\$\$
Continence Care	High	High	High	2	Q2-Q3	\$\$\$
Supply Chain Network	High	Medium	Low	1	Q1-Q2	\$\$
Wound Care	Medium	Medium	Medium	3	Q3-Q4	\$\$
Critical Care	Medium	Low	Medium	4	Q3-Q4	\$

Implementation Roadmap

Phase 1: Foundation (Months 1–3)

- Establish DOE governance structure aligned with HPS lean methodology
- Deploy supply chain network DOEs (inventory buffers, SAP parameter tuning, kanban calibration)
- Initiate ostomy care CeraPlus barrier formulation and adhesive cure DOEs
- Integrate DOE data collection with SAP ERP quality management module
- Train cross-functional teams at Stuarts Draft and Ballina on DOE methodology

Phase 2: Expansion (Months 4–6)

- Launch continence care catheter coating and VaPro tip geometry experiments
- Execute multi-site capacity allocation and distribution routing DOEs
- Begin regulatory pre-submission consultations for product-modifying experiments
- Establish DOE results repository and knowledge management system

Phase 3: Optimization (Months 7–12)

- Deploy wound care and critical care DOEs
- Implement confirmation runs and validated optimal settings
- Build predictive models from DOE results feeding into digital twin capabilities
- Quantify cumulative business value (target: 15–20% supply chain cost reduction)
- Transition to continuous improvement cycle using DOE as standard HPS practice

DOE Governance and Integration Points

Effective DOE execution requires integration with Hollister's existing operational and governance frameworks:

- **HPS Integration** – DOE experiments should be governed as HPS improvement projects with standardized A3 problem-solving documentation and PDCA cycle alignment
- **SAP ERP Data Layer** – Experimental parameters and results should feed into SAP Quality Management (QM) and Production Planning (PP) modules for automated parameter adjustment
- **Regulatory Alignment** – Product-modifying DOEs (especially ostomy barrier and catheter experiments) require Design History File (DHF) documentation per FDA 21 CFR 820 and EU MDR 2017/745
- **Cross-Functional Ownership** – Each DOE should have a designated owner spanning R&D, Manufacturing, Supply Chain, and Quality, with executive sponsorship from Global Operations leadership
- **Knowledge Capture** – DOE results should be encoded into a structured knowledge repository (semantic layer) enabling reuse across sites and future product development programs

Expected Outcomes

When fully implemented, this DOE framework is projected to deliver the following quantified benefits across Hollister's supply chain and product portfolio:

1. 15–20% reduction in supply chain operating costs through inventory buffer optimization and logistics network redesign
2. 10–15% improvement in OEE across manufacturing sites through process parameter optimization
3. Measurable improvement in patient outcomes (barrier wear time, UTI reduction, wound healing rates)
4. 30–40% reduction in expedited orders through improved demand-supply alignment
5. SKU rationalization enabling 10–15% inventory reduction while maintaining service levels
6. Accelerated time-to-market for new product variants through systematic experimental learning
7. Foundation for digital twin and predictive analytics capabilities leveraging DOE-derived models